A (Scalable) Minimalist Algorithm for Shape-Changing Robots in Dense Swarms

Alexandra Nilles, Henry Boekhoff Department of Computer Science, Western Washington University Heather Jin Hee Kim, Julian Prieto Sibley School of Mechanical and Aerospace Engineering, Cornell University Elizabeth Garner, Kirstin Petersen Department of Electrical and Computer Engineering, Cornell University

Resource-Constrained Shape-Changing Robots

Motivation: Shape-Changing Micro-Robots

- Mechanisms include magnetism, bistable mechanisms, electrofluidics, many others

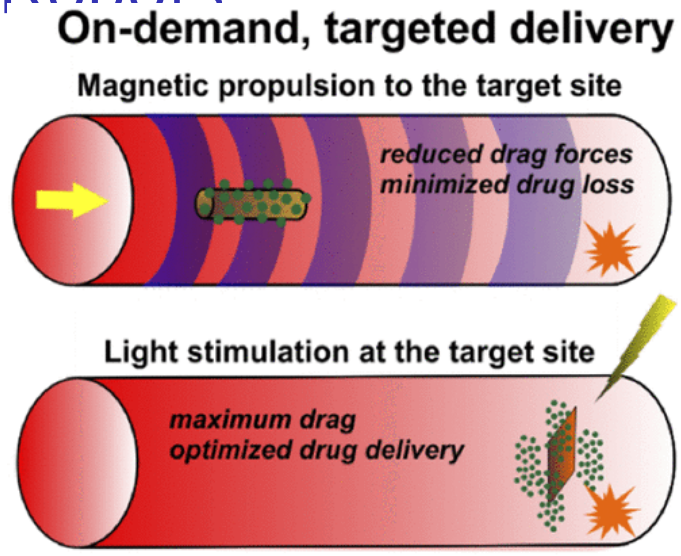


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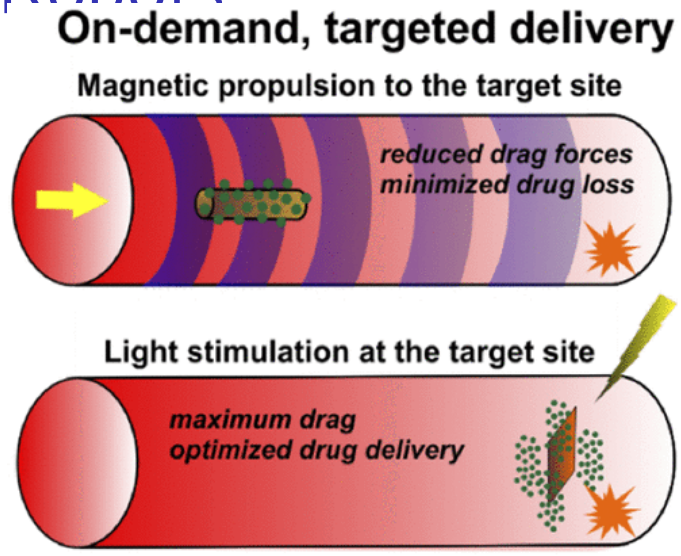


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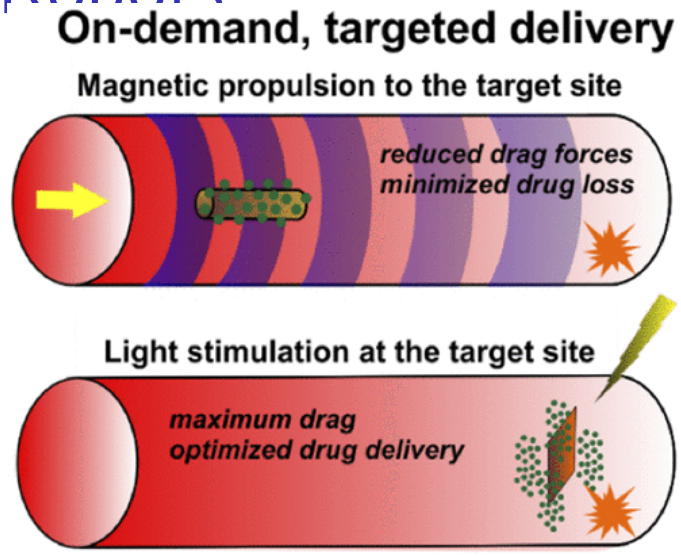


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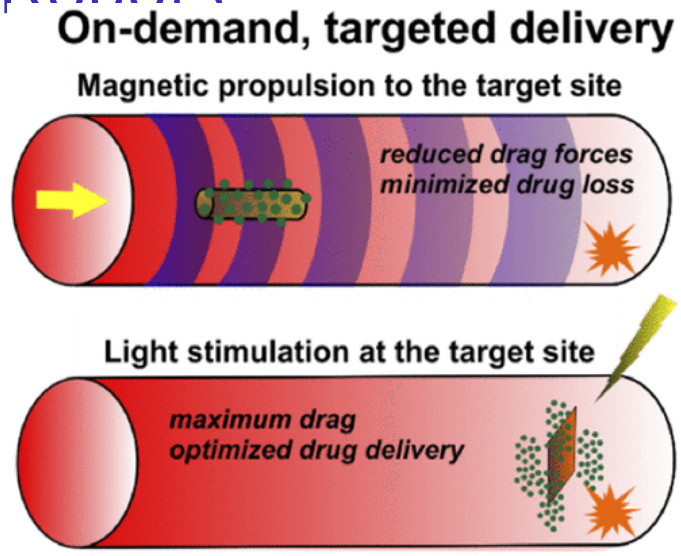


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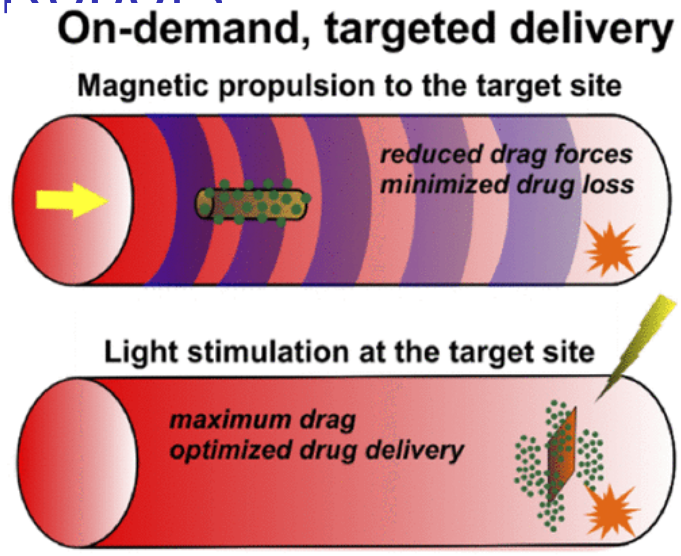


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- Why change shape?
  - Drug delivery
  - Collective dynamics (align, aggregate)

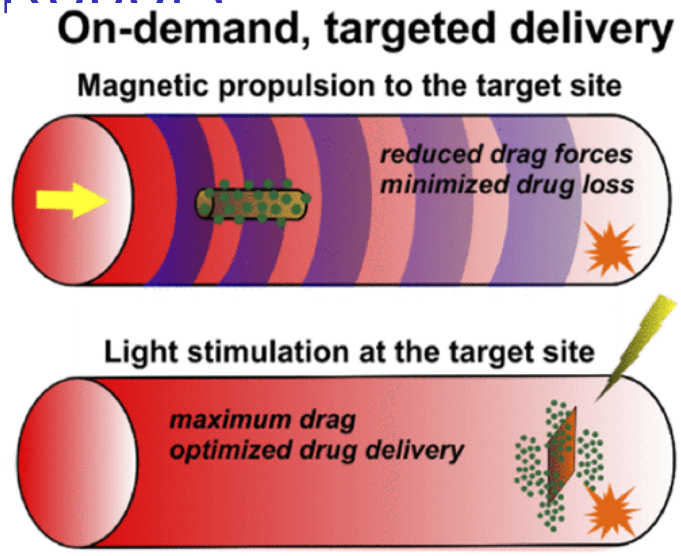


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Motivation: Quorum Sensing without Deliberation

- Externalize computation of density estimation to random walk

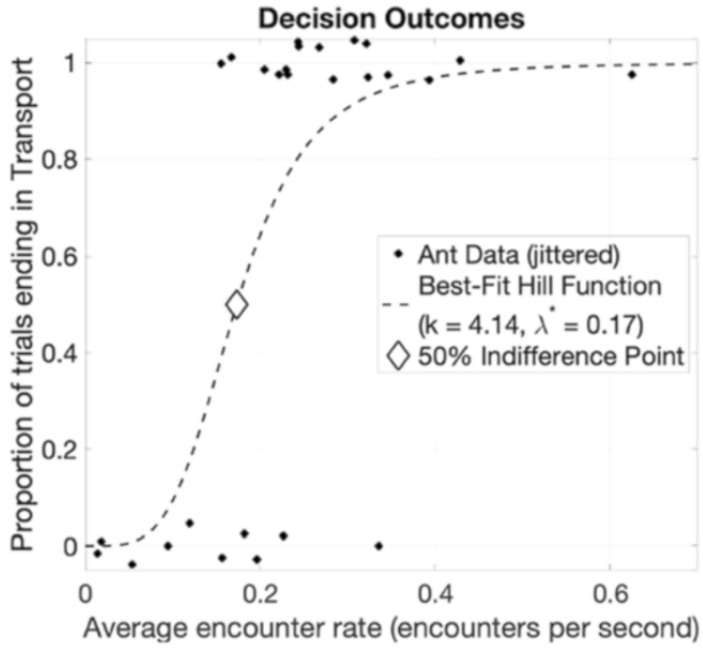


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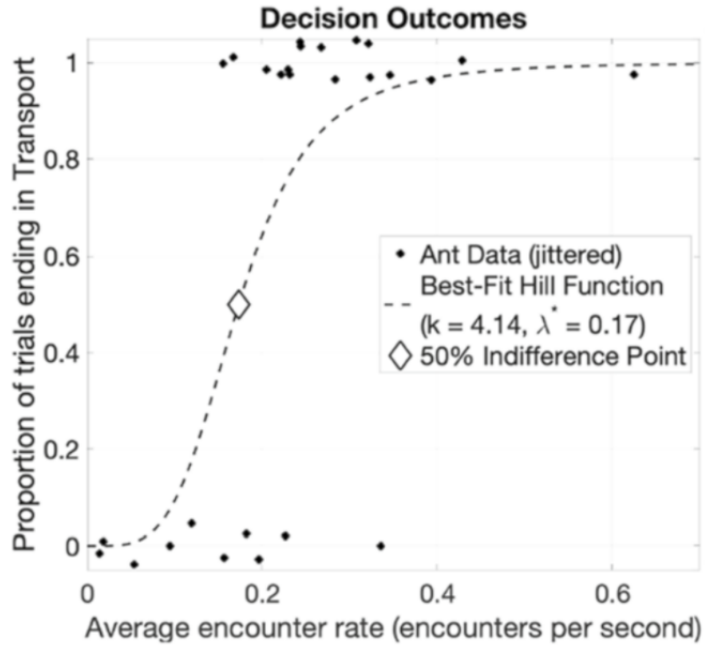


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Motivation: Quorum Sensing without Deliberation

- Externalize computation of density estimation to random walk
- Strong evidence of effective strategies in biological systems
- Minimizing onboard resources: no pheromones or explicit communication

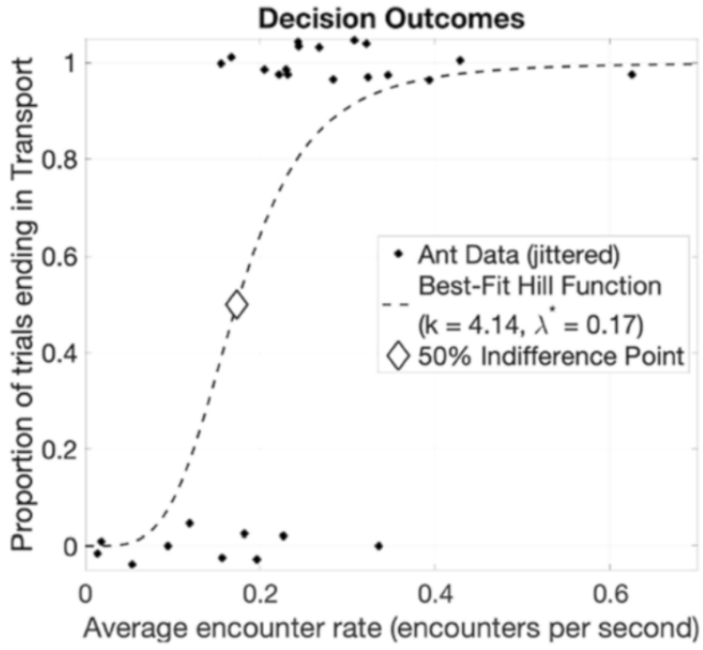


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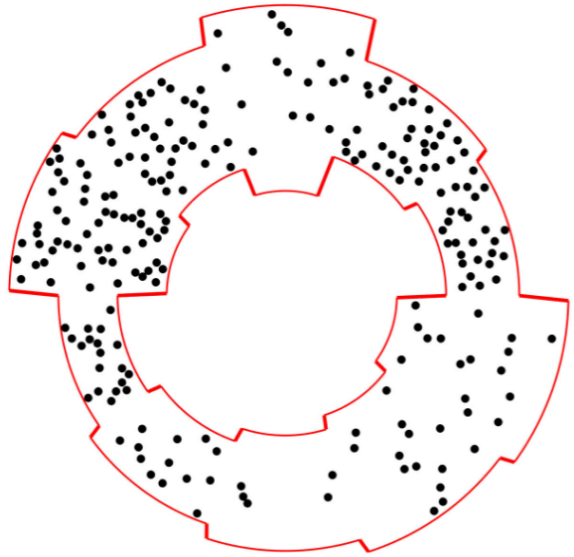


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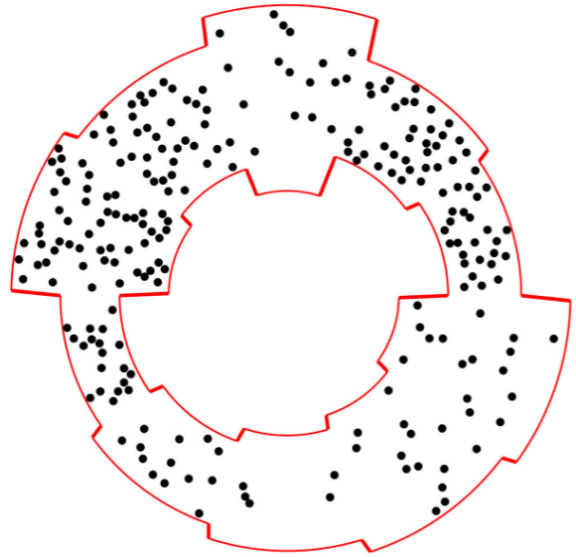


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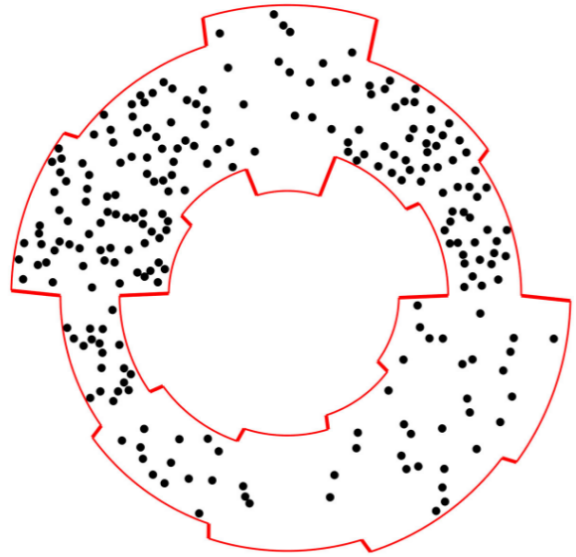


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- Measuring aggregation: Motility induced phase separation (MIPS)

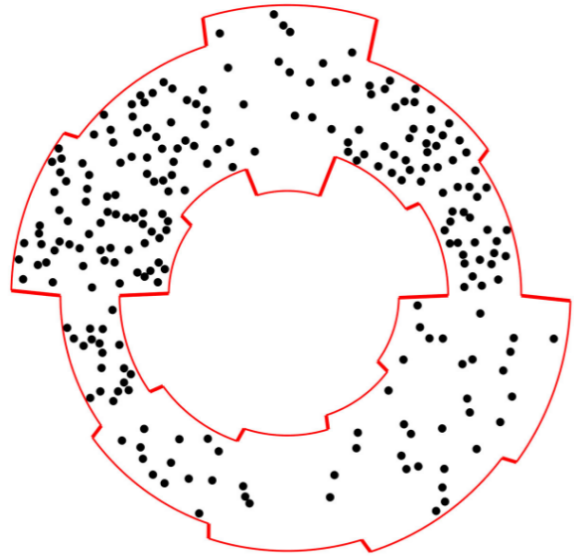


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Our Scenario

Assumptions

- "Wild body" motion: no explicit motion model

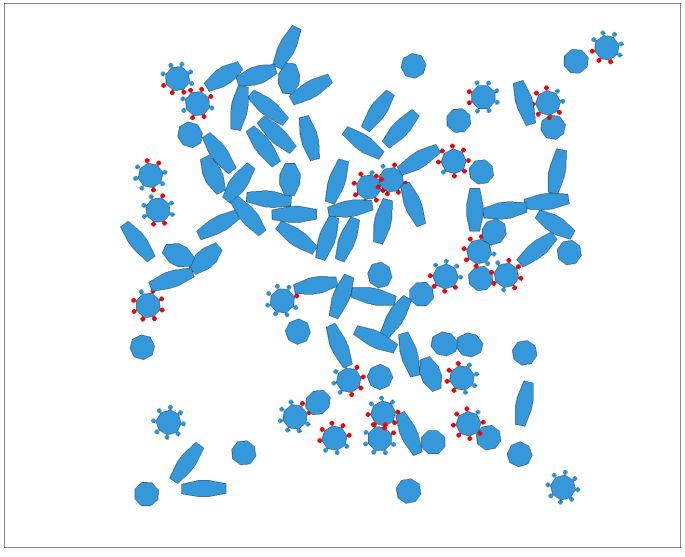


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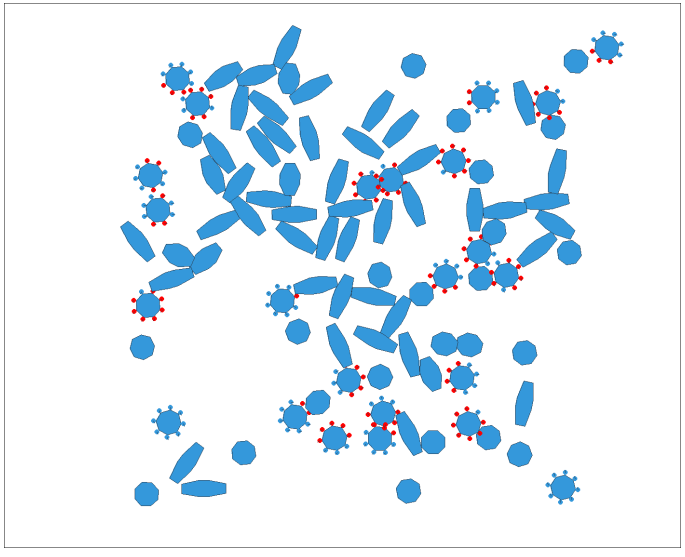


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- Agents will eventually explore everywhere (2D bounded environment)
- Robots have two possible shapes, can change shape individually

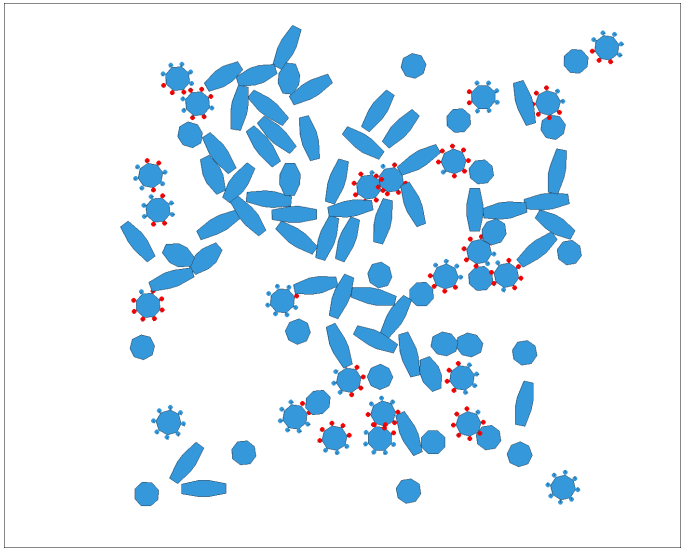


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Research Questions

Can we control the proportion of the collective in each shape using local decisionmaking?

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What effect does shape have on collective dynamics?

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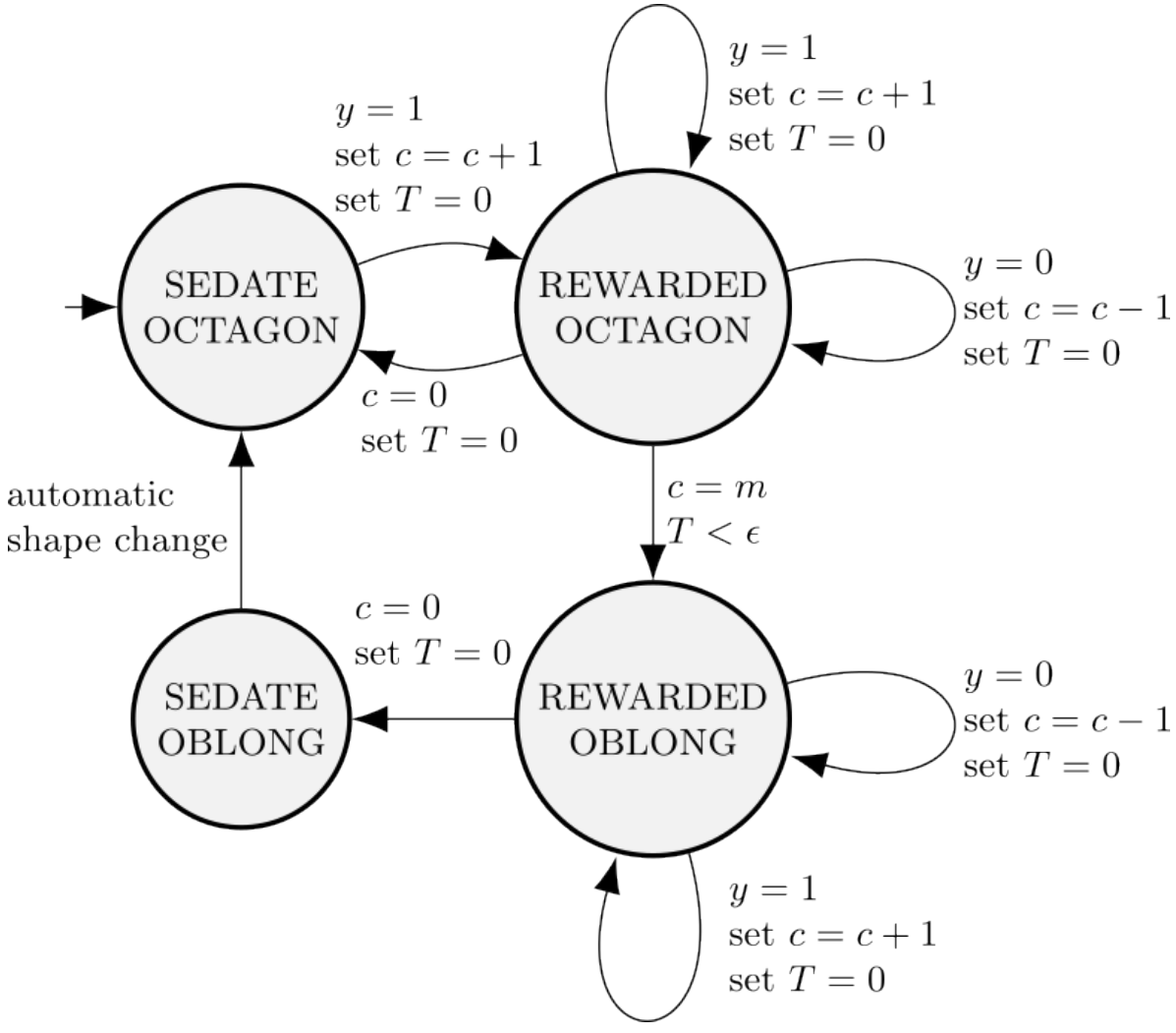
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- Change back when counter reaches 0 (fully sedate)

Proposed State Machine



Resources Required

- Collision, proximity, or contact sensor(s)

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- Two binary state variables

Analysis

Population Dynamics

$$\dot{R} = S\lambda - R/\tau$$
$$\dot{S} = R/\tau - S\lambda$$

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- λ is transition rate from shape 1 to shape 2
- τ is related to transition rate back to shape 1

Probability of Collision

- To determine transition rate, need to understand probability of collision

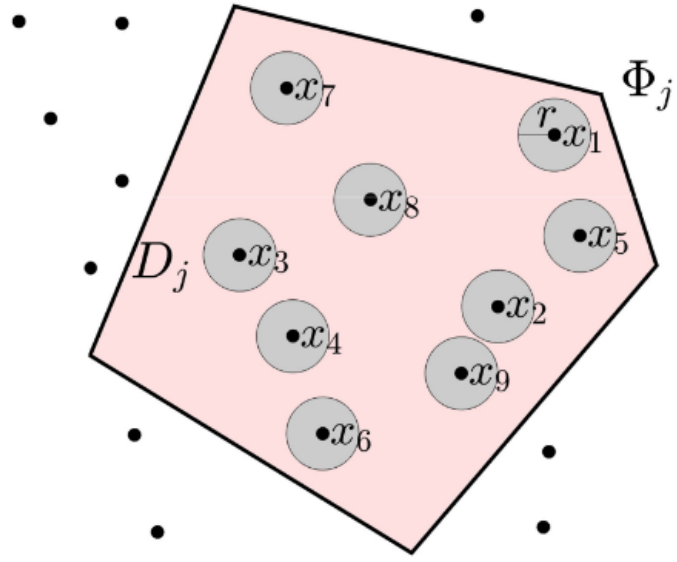


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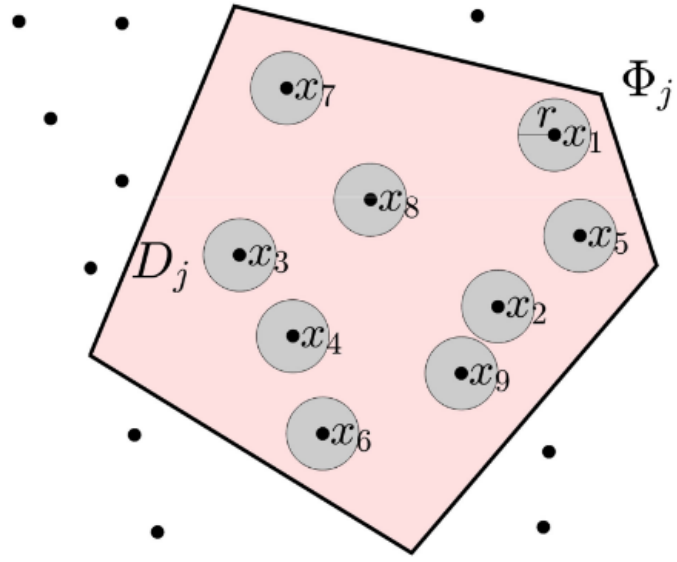


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- Well-mixed (AKA mean field) assumption allows use of Poisson point process

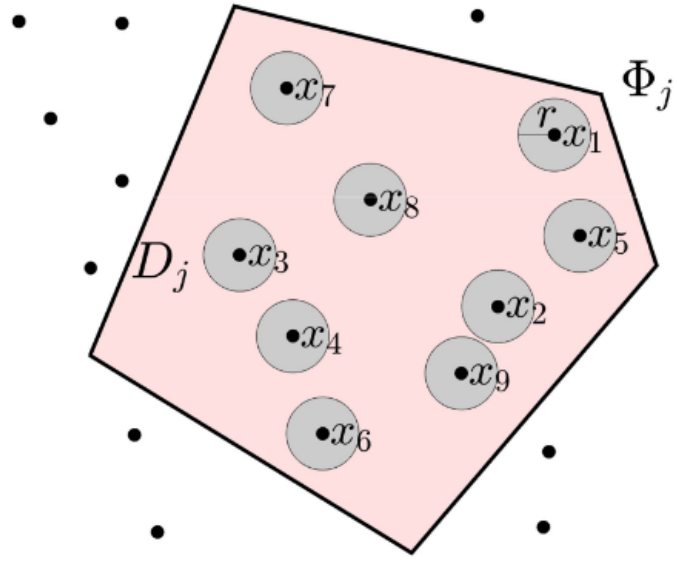


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Density Predictions

Probability of collisions gives:

1. λ: probability of m sequential collisions with less than ϵ elapsed time between them

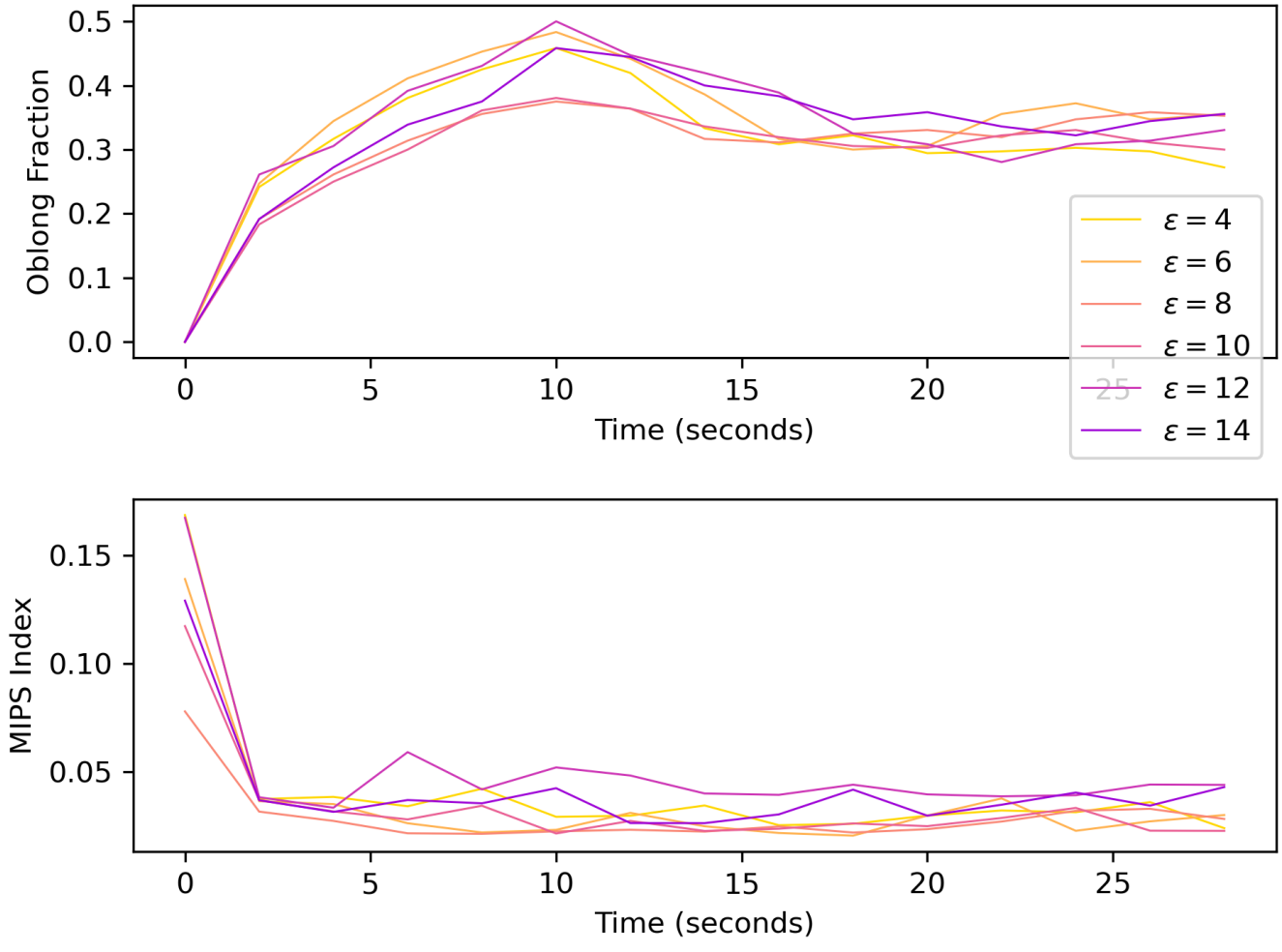
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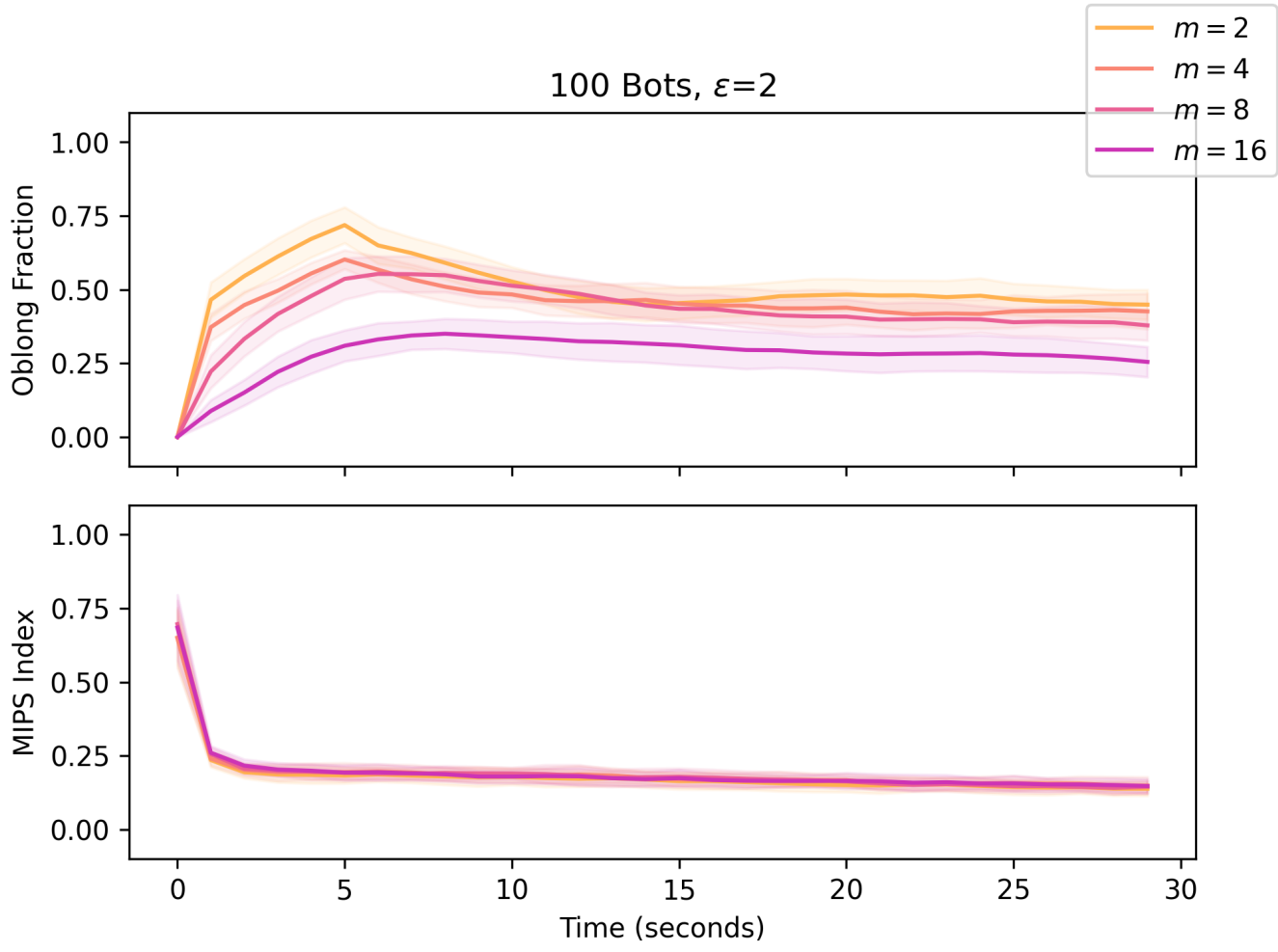
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2. τ: probability of no collisions within $m\epsilon$ time

Results

Low Sensitivity to Timer Threshold ϵ in Dense Settings



Effect of Counter Threshold m



High Sensitivity to Density with Fixed m and ϵ

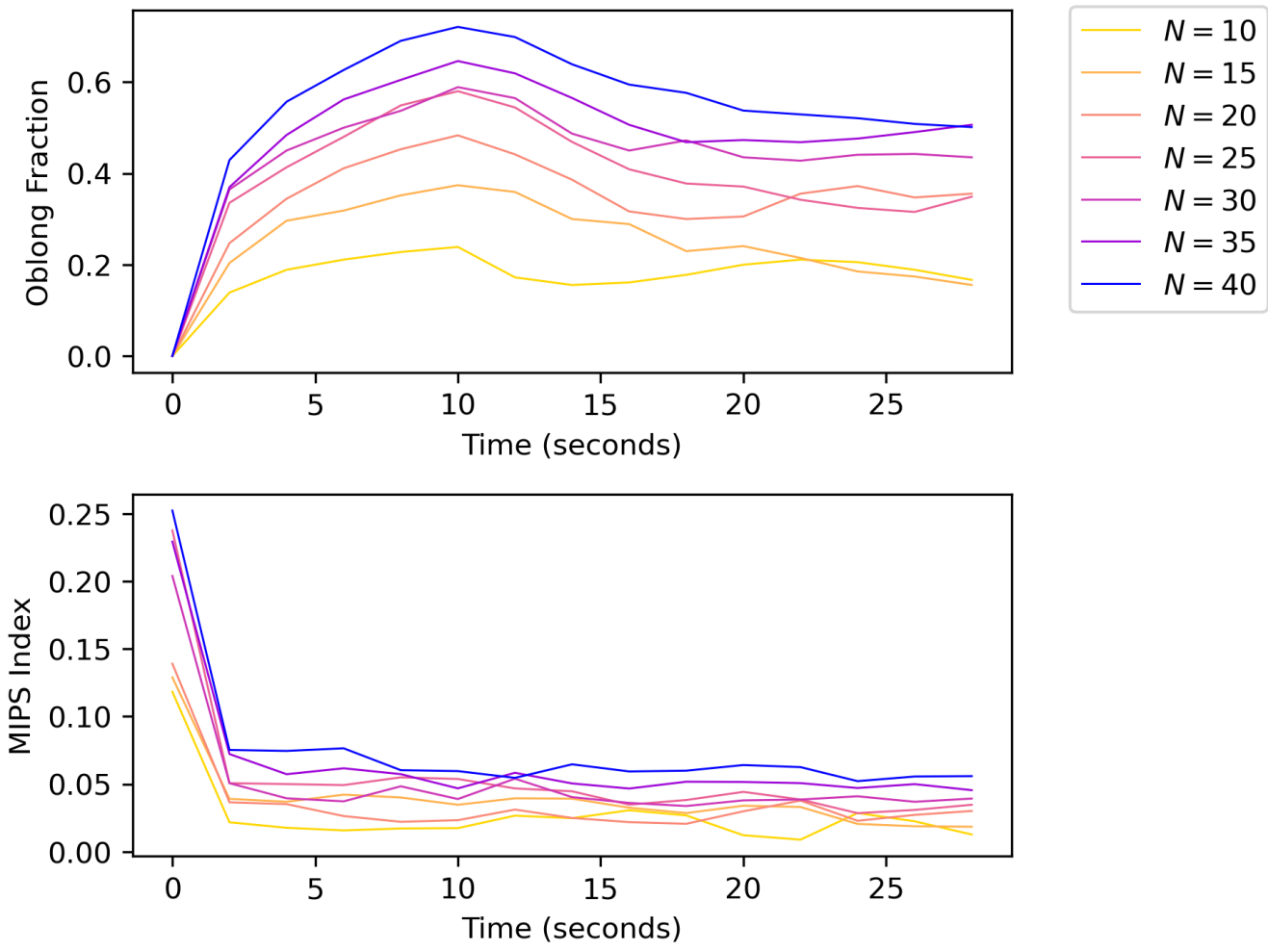


Figure 6: Distributed adaptation to density changes!

Demonstration

Dirty Laundry and Next Steps

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- MIPS normalization and discrete computation



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- Radius assumptions



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- Radius assumptions
- Hardware experiments inconclusive
  - Weaselball fail
  - Micro-robot physics vary widely



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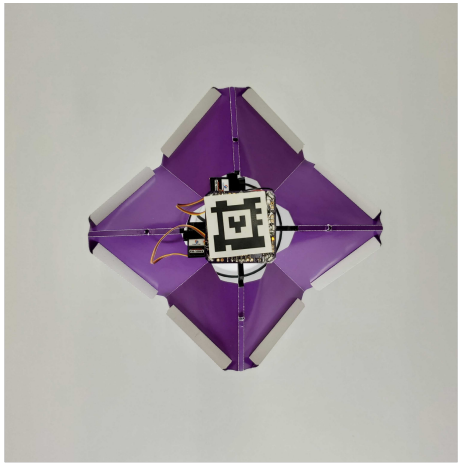
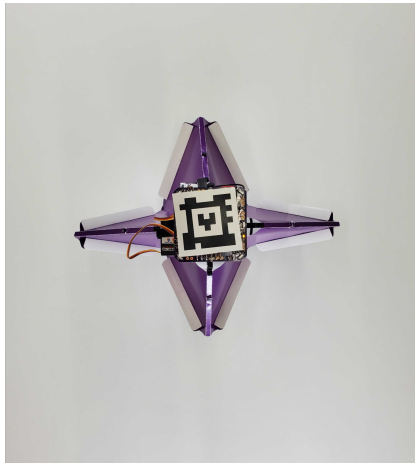
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- Hints at fun problem structure to improve theory

Thank you! Questions?

New lab website: https://mrrp.cs.wwu.edu